

Oizys

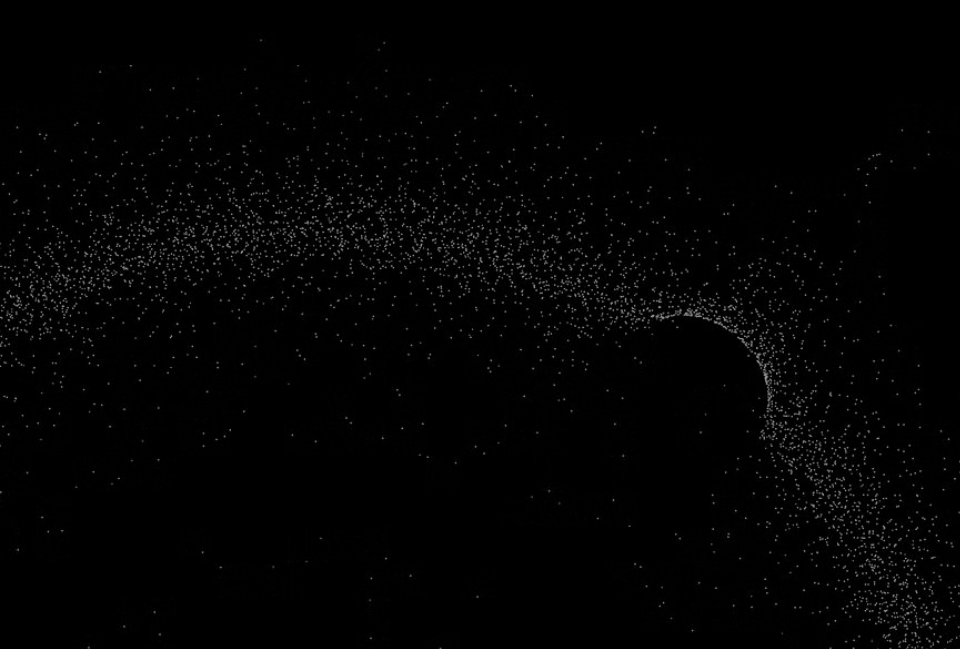
Stress
Bracelet



Karis

Lillian

Content



1 Topic and Importance



2 Problem



3 Solution



4 Overview of Technology



5 Demo!



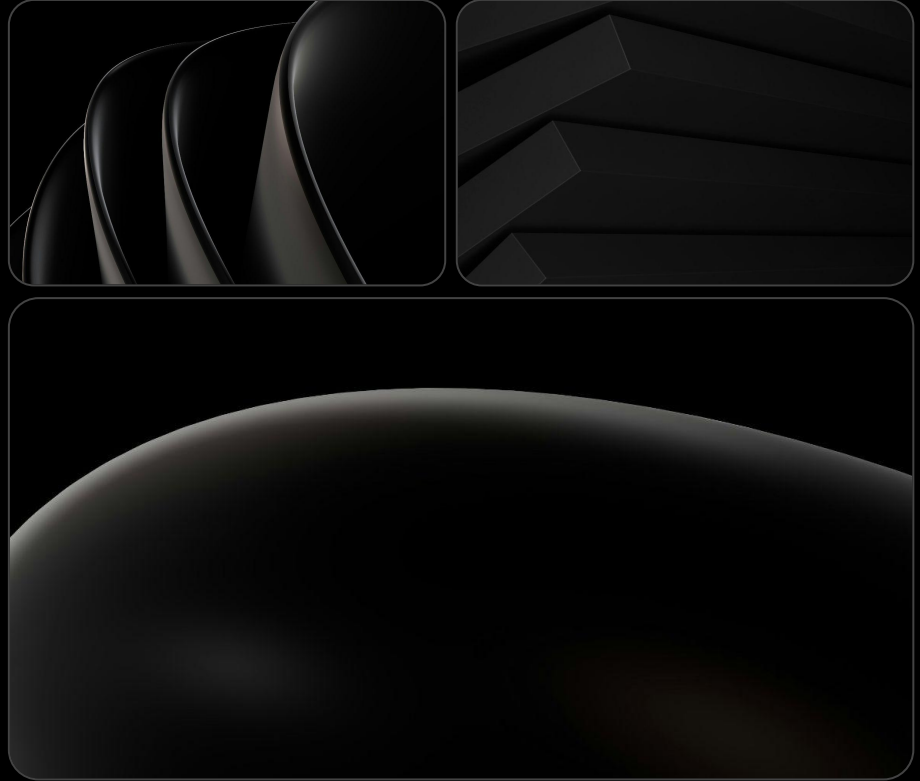
6 Looking Forward



Stress

"Nearly half of all U.S. adults (49%) say that stress has negatively affected their behavior."

– American Psychological Association



"As of August 2024, 31% of adults surveyed worldwide thought stress was the biggest health problem in their country"

– Statista

The Stress Cycle

Stress Starts

Step 1

Internal and external factors cause stress to an individual.

Short term stress can be helpful, but long term stress is extremely harmful.

Environment

Step 2

People in a competitive and cutthroat environment **seeing high stress as normal**, making it harder to identify or address.

Normalize Stressed

Step 3

Normalized or get used to it, so don't seek help and let it persist.

If everyone is stressed, I should be too!

Subtle Symptoms

Step 4

Symptoms often show up in subtle ways (fatigue, irritability, or muscle tension) that people don't recognize as stress.

But this is **detrimental to long term health**.

(break) The Stress Cycle

Stress Starts

Step 1

Internal and external factors cause stress to an individual.

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Step 2

People in a competitive and cutthroat seeing high stress as normal, making it harder to identify or address.

Normalize Stressed

Step 3

Stress is normalized as a normal part of life. People see stress as a necessary part of life. If stress is normalized, it becomes a part of the culture.



Subtle Symptoms

Step 4

Stress is normalized as a normal part of life. People see stress as a necessary part of life. If stress is normalized, it becomes a part of the culture. But this is not good for long term health.



Current Solutions

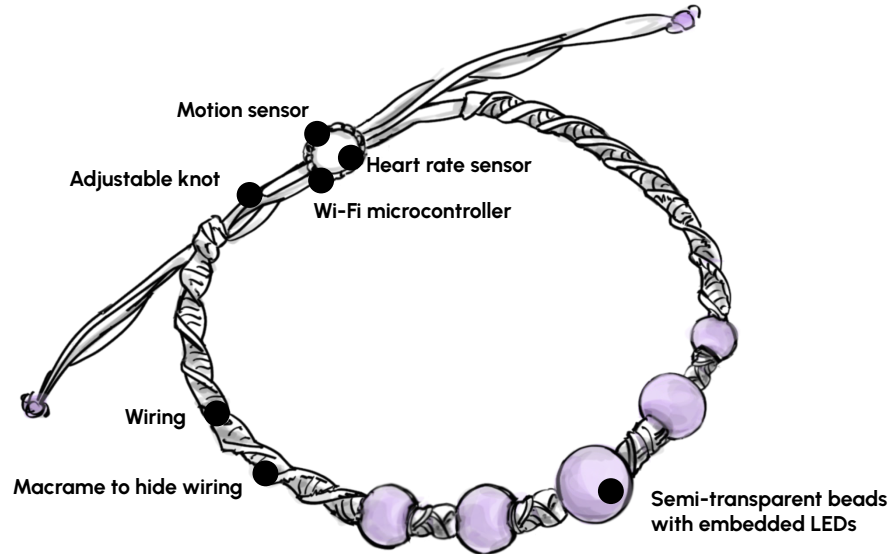
- Smart watches that just display a lot of data that is often unnecessary
 - Bulky and Flashy
 - Lot of raw data -> Cognitive Overload
- Sometimes more choice causes more stress



> We want to make something that makes stress data digestible for people



Final Concept



Stress Bracelet

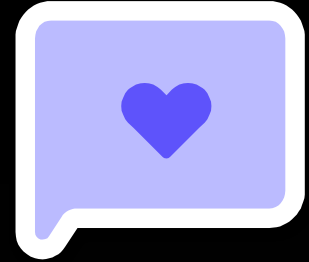
Ambient device to signal users when they are stressed

Takes heart rate and motion metrics from sensors to determine whether or not high levels of stress are present

Determined stress level translates into subtle glow to indicate high levels → help users be aware

Why Heart Rate and Movement?

- Many Studies including one from the National Institute of Health have shown that Heart Rate Variability is a strong indicator of Stress
 - <https://pmc.ncbi.nlm.nih.gov/articles/PMC5900369/>
- However, physical activity is often an indicator of a elevated heart rate
 - By accounting for both, we minimize our **false positives**



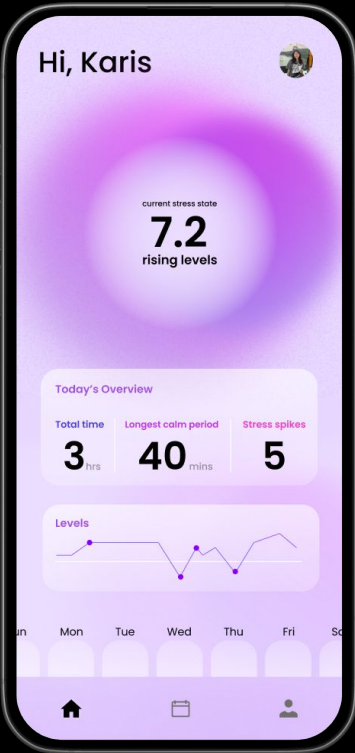


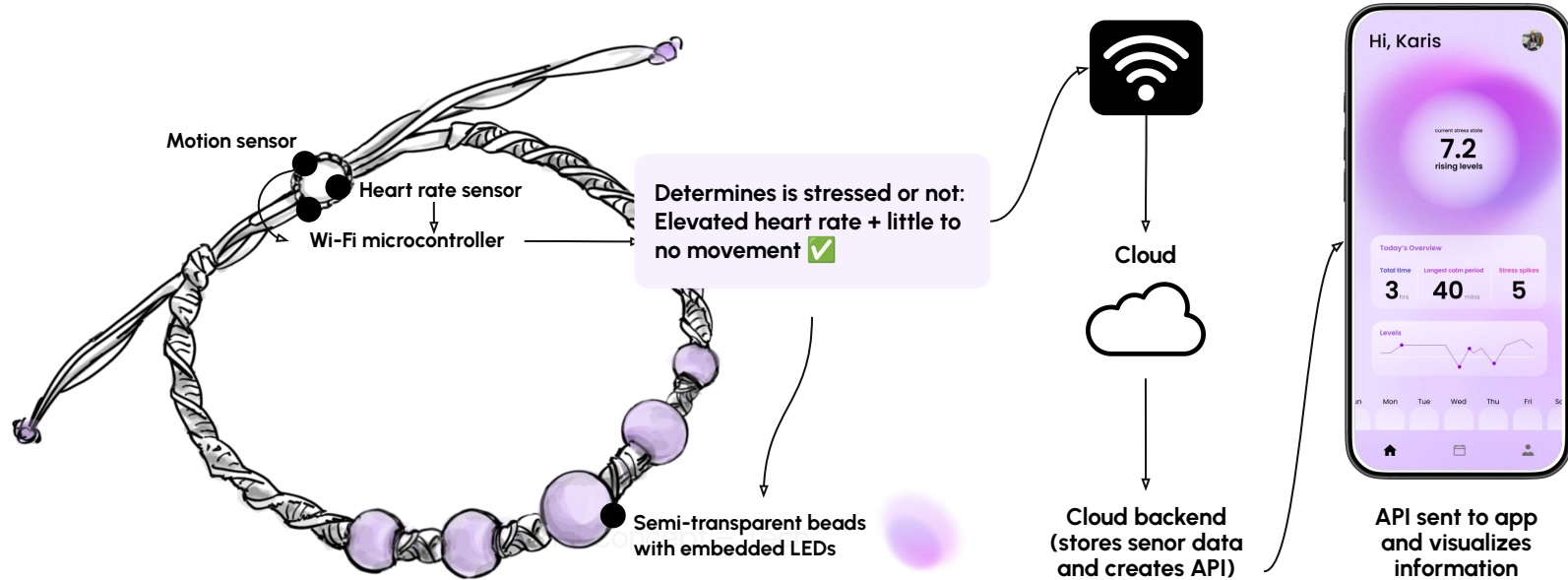
Not Stressed



Stressed

Final Concept – Part 2





Our Solution

1 Subtle



Many people dislike smart watches because they look flashy or bulky

2 Intuitive & Simple



Digestible and easy to see/understand at a glance

Form that mimics existing accessories that people are already familiar with.

3 Trackable Information



External app displays insights about stress levels and duration

No need for user logs or calls

4 Integrates Course Content



Ambient device – function

Enchantment – wearable has no interface

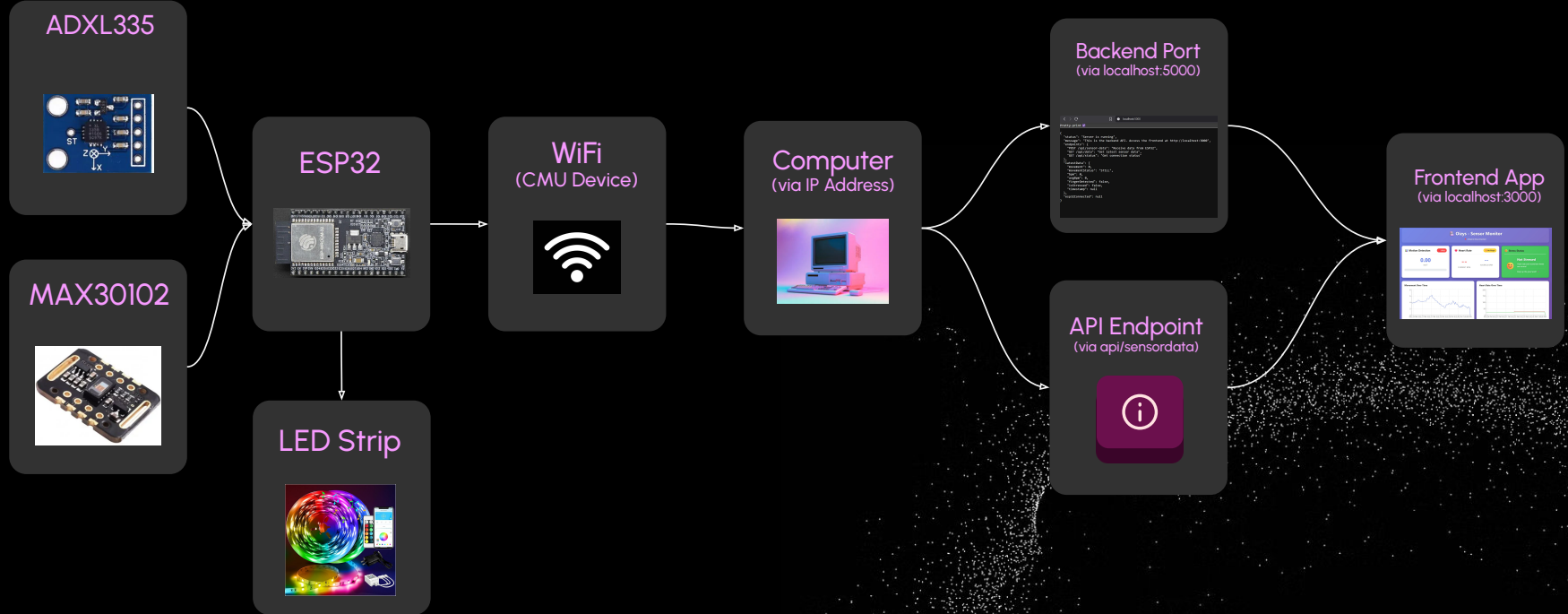
Aesthetic – form

Why does this solve the Problem?

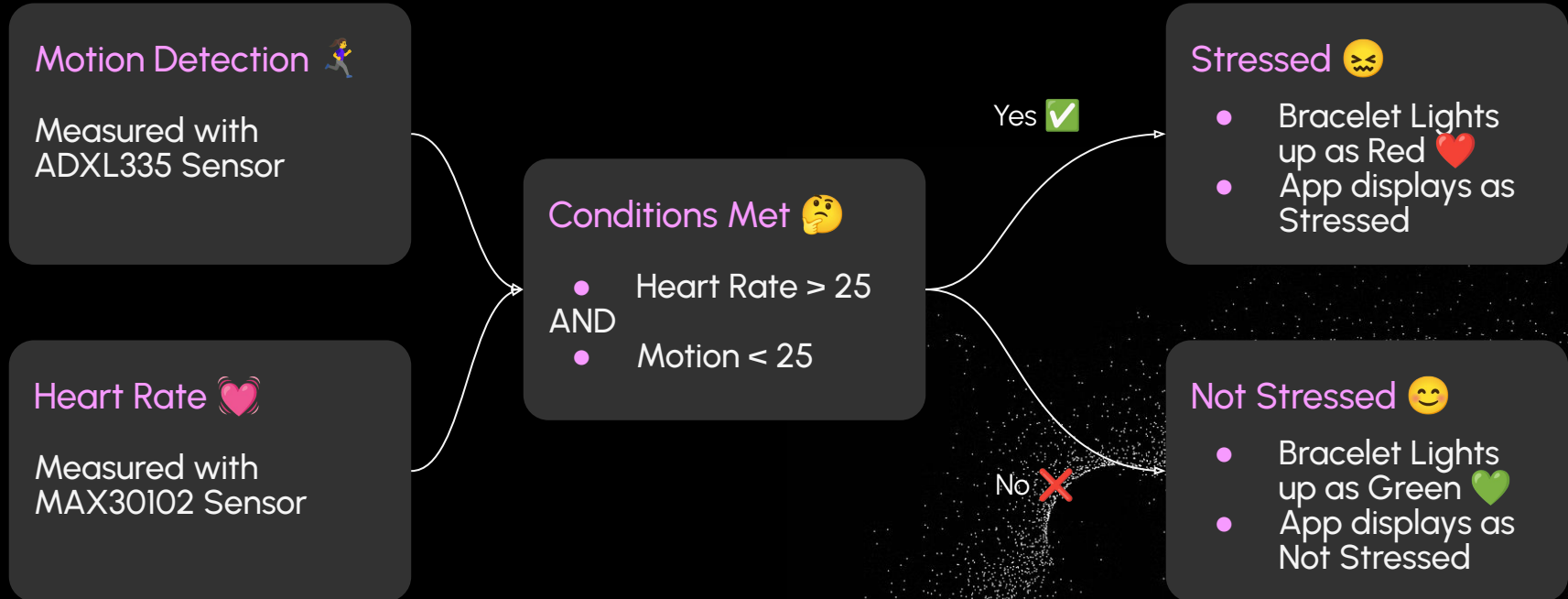
- ✓ Targets unawareness or normalization issues through a nudge, increasing stress intervention initiatives

Prototype

Tech Overview | Data Flow



How Stress is Determined

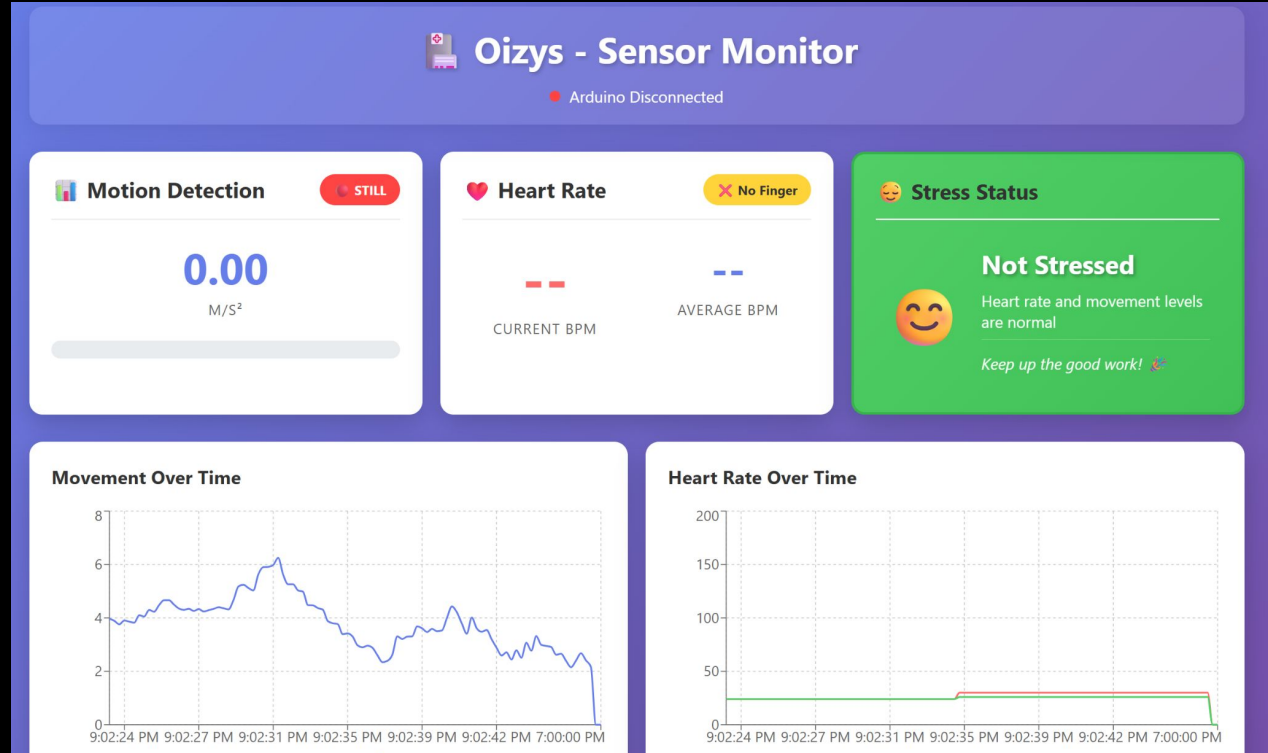


App Interface

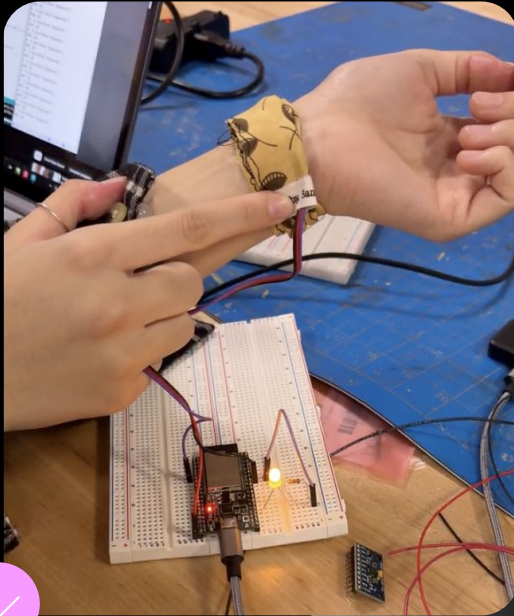


Built in a React
Framework

(Vibe Coded with Cursor AI)



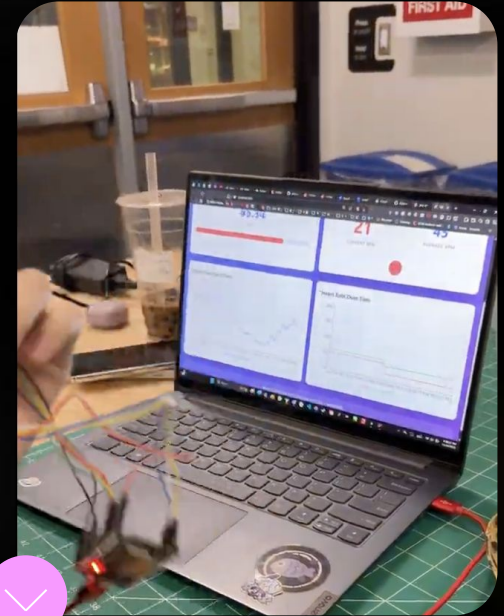
Process



Testing Heart Rate Sensor

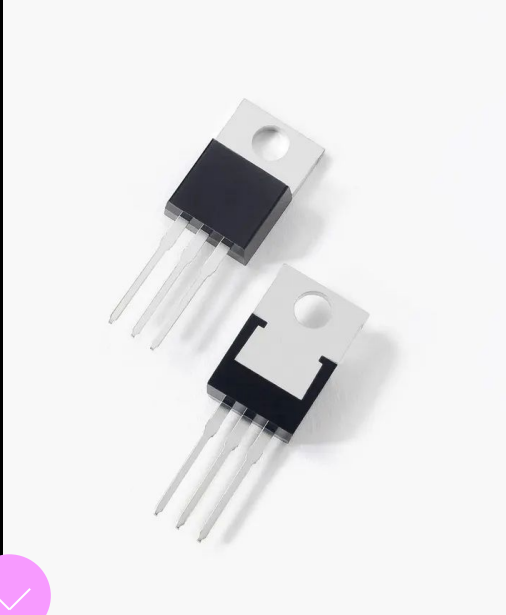


Getting Accelerometer to work



Connecting data to app

Process



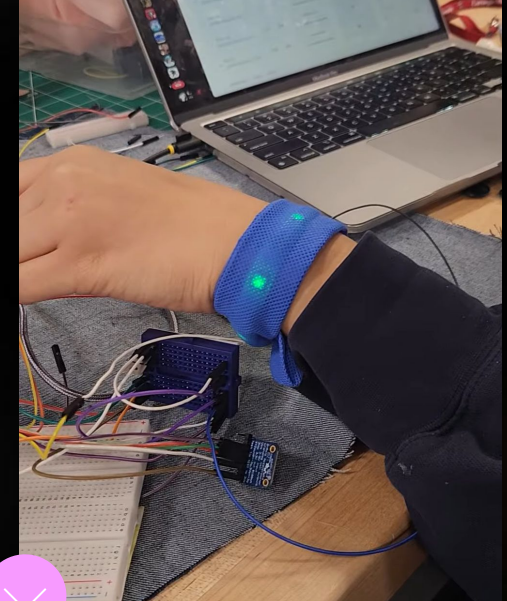
Add External Power Source

```
localhost:5000
Pretty-print ✓

{
  "status": "Server is running",
  "message": "This is the backend API. Access the frontend at http://localhost:3000",
  "endpoints": {
    "POST /api/sensor-data": "Receive data from ESP32",
    "GET /api/data": "Get latest sensor data",
    "GET /api/status": "Get connection status"
  },
  "latestData": {
    "movement": 0,
    "movementStatus": "STILL",
    "bpm": 0,
    "avgBpm": 0,
    "fingerDetected": false,
    "isStressed": false,
    "timestamp": null
  },
  "esp32Connected": null
}
```

```
[0] [9:02:41 PM] POST /api/sensor-data from 172.26.203.0
[0] [9:02:41 PM] POST /api/sensor-data from 172.26.203.0
[0] [9:02:42 PM] POST /api/sensor-data from 172.26.203.0
[0] [9:02:42 PM] POST /api/sensor-data from 172.26.203.0
[0] [9:02:42 PM] POST /api/sensor-data from 172.26.203.0
[0] [9:02:42 PM] POST /api/sensor-data from 172.26.203.0
[0] [9:02:42 PM] POST /api/sensor-data from 172.26.203.0
[0] [9:02:42 PM] POST /api/sensor-data from 172.26.203.0
[0] [9:02:43 PM] POST /api/sensor-data from 172.26.203.0
```

Wirelessly Connect Data



Displaying it in the LEDs

Challenges, Learning, & Future Steps

Inaccurate Heart Rate Sensor

Our Heart Rate sensor had issues detecting and calibrating heart rate.

- We switched our sensor and adjusted our calculation method for BPM

Power Source

Our remote power source was too strong and was shorting electronics.

- We worked with professors and the TA to troubleshoot our chip and added a transistor

Housing and Aesthetics

Our prototype housing is currently not ideal. Sensors too large to fit inside a wearable

- Want to adjust our housing and sensors in the future to be more sleek and compact

HRV Calculation

Currently our app and website only calculates BPM, not long term HRV

- Add a HRV calibration feature to see the user's normal BPM and make tracking more visible on app

DEMO

Let's jump into
showing the Oizys
Stress Bracelet!



Let's work together

Video Demo



THANK YOU



Questions/
comments?